

Oscar Projekt

SAPunanje mačora

2025-11-02

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```
library(dplyr)
library(readr)
library(ggplot2)
library(tidyverse)
library(boot)
library(gridExtra)
library(car)
library(tinytex)

oscars_dataset <- read_delim("oscars_dataset.csv", delim = ";", escape_double = FALSE, trim_ws = TRUE)
```

Pregled naših podataka

```
##      Unnamed: 0      Film      Oscar Year      Film Studio/Producer(s)
##  Min.   : 0.0    Length:571    Length:571    Length:571
##  1st Qu.:142.5  Class :character  Class :character  Class :character
##  Median :285.0  Mode  :character  Mode  :character  Mode  :character
##  Mean   :285.0
##  3rd Qu.:427.5
##  Max.   :570.0
##
##      Award      Year of Release  Movie Time  Movie Genre
##  Length:571    Min.   :1927    Min.   : 66.0    Length:571
##  Class :character  1st Qu.:1944    1st Qu.:107.0    Class :character
##  Mode  :character  Median :1972    Median :121.0    Mode  :character
##                      Mean   :1973    Mean   :124.9
##                      3rd Qu.:2001    3rd Qu.:136.5
##                      Max.   :2021    Max.   :238.0
##
##      Main Genre      Subgenre      IMDB Rating      IMDB Votes
##  Length:571    Length:571    Min.   :5.60    Min.   : 18
##  Class :character  Class :character  1st Qu.:7.30    1st Qu.: 9660
##  Mode  :character  Mode  :character  Median :7.60    Median : 58996
##                      Mean   :7.57    Mean   :208480
##                      3rd Qu.:7.90    3rd Qu.:259429
##                      Max.   :9.30    Max.   :2452594
##
##      Content Rating      Directors      Original Release Date Production Company
```

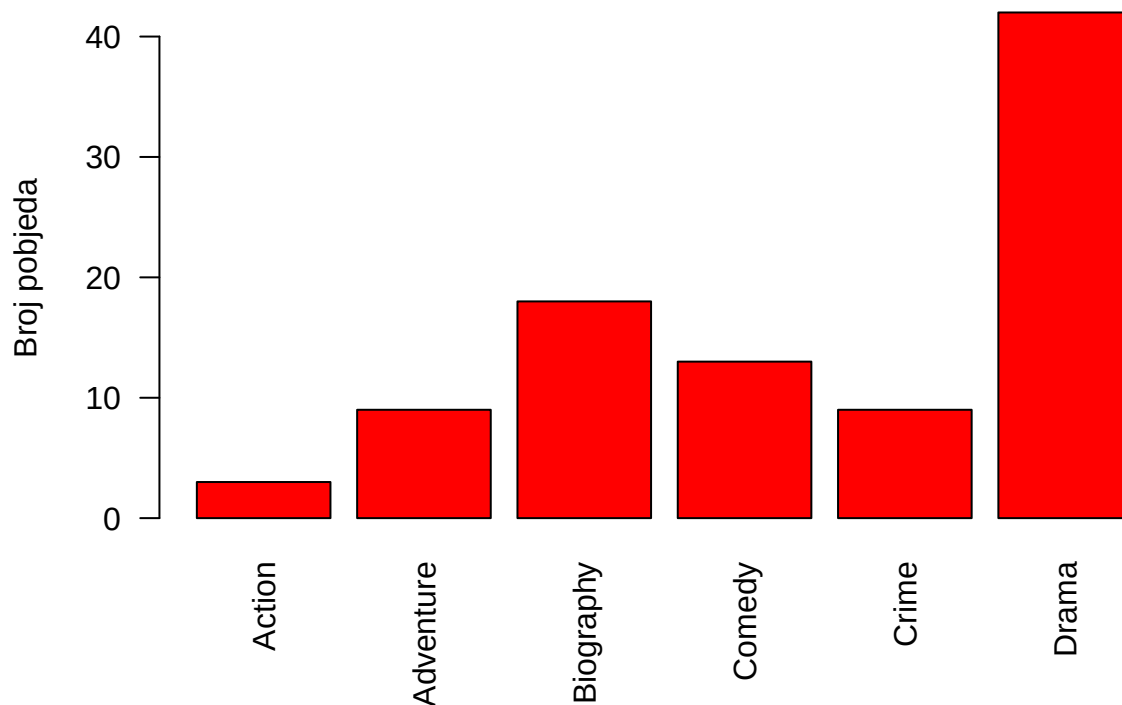
```
## Length:571      Length:571      Min.   :1927-08-12      Length:571
## Class :character Class :character 1st Qu.:1951-11-12      Class :character
## Mode  :character Mode  :character Median :1979-01-01      Mode  :character
##                                     Mean  :1978-05-30
##                                     3rd Qu.:2006-08-31
##                                     Max.   :2021-02-26
##                                     NA's    :112
## Tomatometer Rating Tomatometer Count Audience Rating Audience Count
## Min.   : 29.00      Min.   : 5.0      Min.   :21.00      Min.   : 6
## 1st Qu.: 84.00      1st Qu.: 26.0      1st Qu.:78.00      1st Qu.: 6270
## Median : 91.00      Median : 59.0      Median :85.00      Median : 35290
## Mean   : 87.64      Mean   :124.1      Mean   :82.57      Mean   : 493614
## 3rd Qu.: 95.00      3rd Qu.:217.0      3rd Qu.:90.00      3rd Qu.: 113078
## Max.   :100.00      Max.   :598.0      Max.   :98.00      Max.   :35797635
## NA's    :112        NA's    :112        NA's    :112        NA's    :112
```

#Jesu li među pobjednicima neki žanrovi češći? Hipoteze:

H0 - Svi žanrovi imaju jednaku vjerojatnost pobjede H1 - Postoji razlika u učestalosti pobjeda među žanrovima

alpha = 0.05

Broj pobjednika po žanru



```
## [1] "Kontingencijska tablica (frekvencije):"
```

```
##
```

```
##      Nominee Winner
```

##	Action	36	3
##	Adventure	36	9
##	Animation	3	0
##	Biography	83	18
##	Comedy	95	13
##	Crime	39	9
##	Drama	173	42
##	Family	1	0
##	Film-Noir	2	0
##	Horror	3	0
##	Music	1	0
##	Musical	2	0
##	Romance	1	0
##	Western	2	0

[1] "Kontingencijska tablica za subžanrove (frekvencije):"

##			
##		Nominee	Winner
##	Adventure	20	2
##	Biography	6	2
##	Comedy	1	0
##	Comedy	4	2
##	Crime	13	3
##	Drama	215	40
##	Family	11	4
##	Fantasy	8	0
##	History	12	1
##	Musical	20	3
##	Mystery	13	1
##	Noir	2	2
##	None	43	11
##	Romance	77	15
##	Sci-Fi	4	0
##	Sport	2	2
##	Thriller	8	1
##	War	14	3
##	Western	4	2

[1] "Očekivane frekvencije:"

##			
##		Nominee	Winner
##	Action	32.5796848	6.4203152
##	Adventure	37.5919440	7.4080560
##	Animation	2.5061296	0.4938704
##	Biography	84.3730298	16.6269702
##	Comedy	90.2206655	17.7793345
##	Crime	40.0980736	7.9019264
##	Drama	179.6059545	35.3940455
##	Family	0.8353765	0.1646235
##	Film-Noir	1.6707531	0.3292469
##	Horror	2.5061296	0.4938704

```
## Music      0.8353765  0.1646235
## Musical    1.6707531  0.3292469
## Romance    0.8353765  0.1646235
## Western    1.6707531  0.3292469
```

```
## [1] "Očekivane frekvencije za subžanrove:"
```

```
##
##           Nominee      Winner
## Adventure 18.3782837  3.6217163
## Biography  6.6830123  1.3169877
## Comdey     0.8353765  0.1646235
## Comedy     5.0122592  0.9877408
## Crime      13.3660245  2.6339755
## Drama      213.0210158 41.9789842
## Family     12.5306480  2.4693520
## Fantasy     6.6830123  1.3169877
## History    10.8598949  2.1401051
## Musical    19.2136602  3.7863398
## Mystery    11.6952715  2.3047285
## Noir       3.3415061  0.6584939
## None       45.1103327  8.8896673
## Romance    76.8546410 15.1453590
## Sci-Fi     3.3415061  0.6584939
## Sport      3.3415061  0.6584939
## Thriller   7.5183888  1.4816112
## War        14.2014011  2.7985989
## Western    5.0122592  0.9877408
```

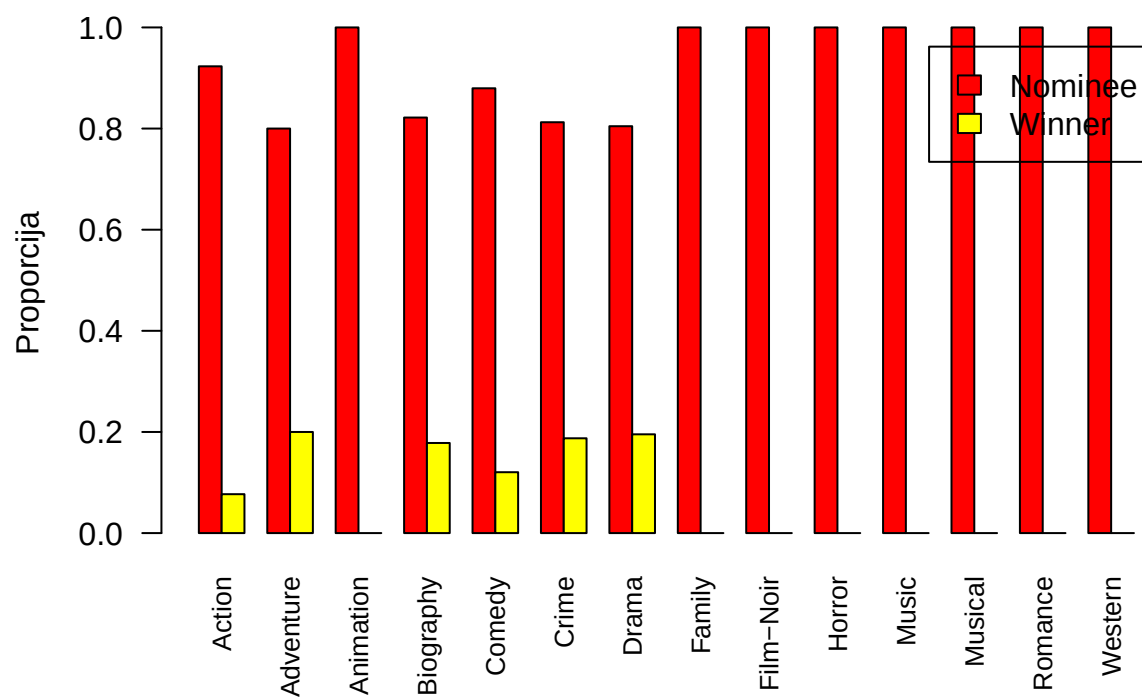
```
## [1] "Rezultat testa za genre: "
```

```
##
## Pearson's Chi-squared test
##
## data:  tablica_zanr_award
## X-squared = 8.8789, df = 13, p-value = 0.782
```

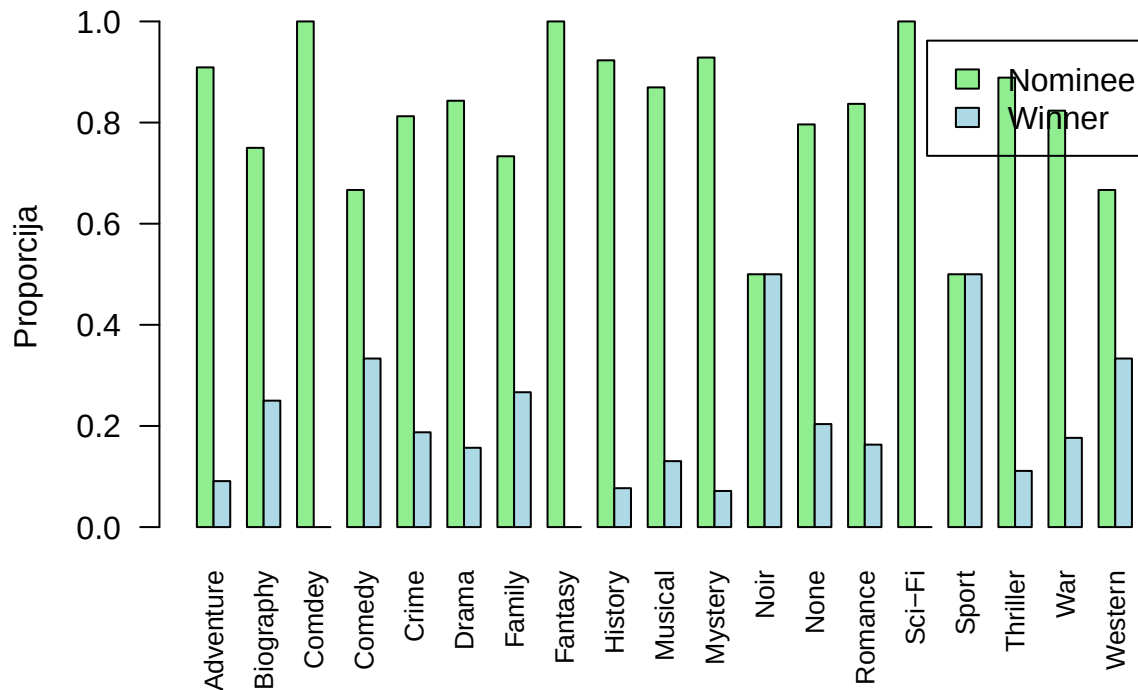
```
## [1] "Rezultat testa za subgenre: "
```

```
##
## Pearson's Chi-squared test
##
## data:  tablica_zanr_award2
## X-squared = 16.803, df = 18, p-value = 0.5367
```

Udio pobjeda i nominacija po žanrovima

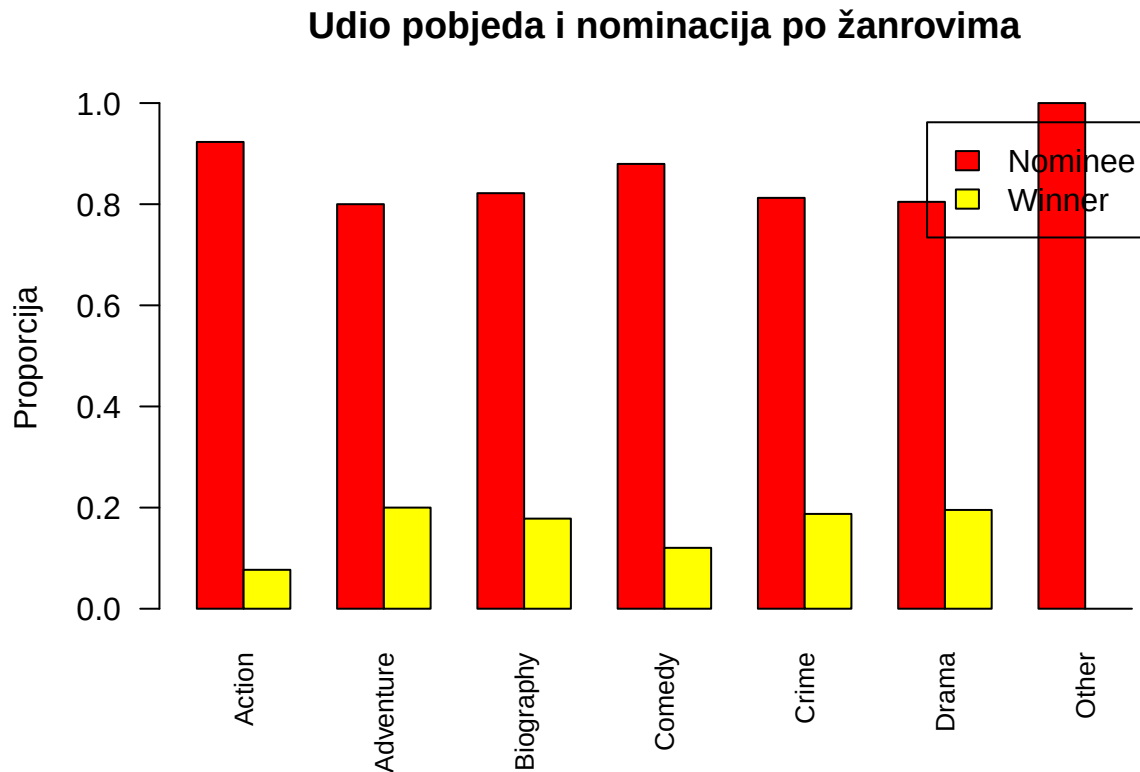


Udio pobjeda i nominacija po subžanrovima



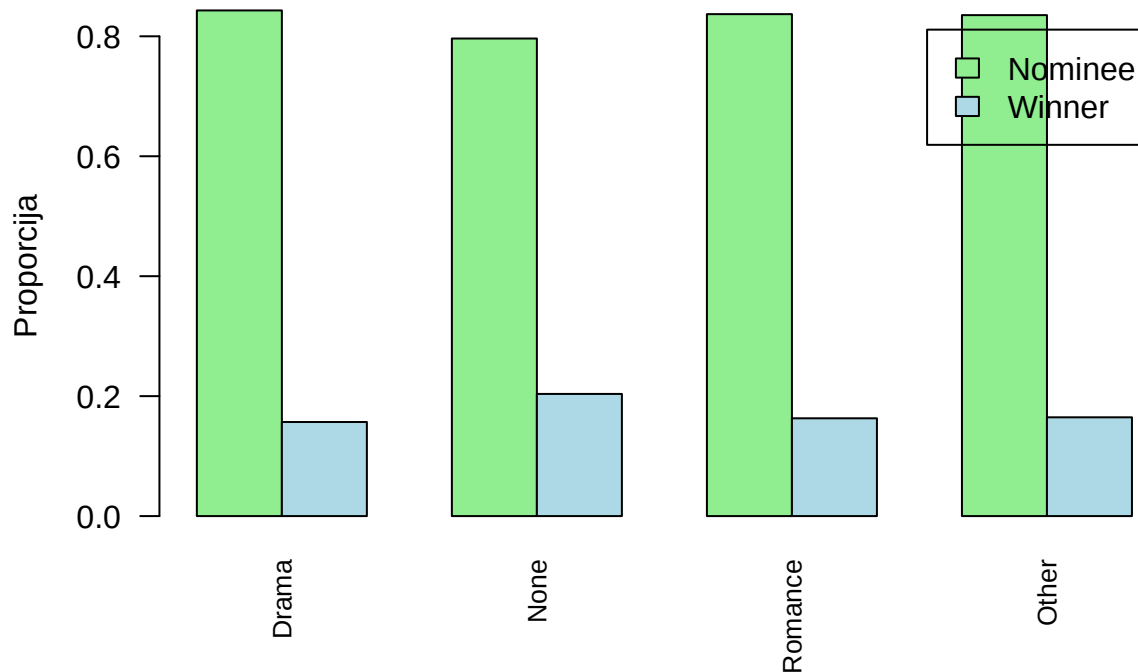
```
## [1] "Grupiranje rijetkih žanrova:"
## [1] "Nova kontingencijska tablica (grupirano):"
##
##      Nominee Winner
## Action      36      3
## Adventure   36      9
## Biography   83     18
## Comedy     95     13
## Crime       39      9
## Drama      173     42
## Other       15      0
## [1] "Očekivane frekvencije nakon grupiranja:"
##
##      Nominee  Winner
## Action  32.57968  6.420315
## Adventure 37.59194  7.408056
## Biography 84.37303 16.626970
## Comedy   90.22067 17.779335
## Crime     40.09807  7.901926
## Drama    179.60595 35.394046
## Other     12.53065  2.469352
##
## Pearson's Chi-squared test with simulated p-value (based on 2000
## replicates)
##
## data:  tablica_finalna
```

```
## X-squared = 8.8789, df = NA, p-value = 0.1824
```



```
## [1] "Grupiranje rijetkih subžanrova:"
## [1] "Subžanrovi s manje od 31 ukupnih filmova idu u 'Other'."
## [1] "Nova kontingencijska tablica (grupirano s pragom):"
##
##           Nominee Winner
## Drama         215     40
## None           43     11
## Romance        77     15
## Other         142     28
## [1] "Očekivane frekvencije nakon grupiranja (Sve bi trebale biti >= 5):"
##
##           Nominee  Winner
## Drama  213.02102 41.978984
## None   45.11033  8.889667
## Romance 76.85464 15.145359
## Other  142.01401 27.985989
##
## Pearson's Chi-squared test
##
## data:  tablica_finalna2
## X-squared = 0.71306, df = 3, p-value = 0.8701
```

Udio pobjeda po subžanrovima (Korigirano)



Na temelju originalnog testa H_0 odbacujemo, što znači da prihvaćamo H_1 , dok za test s grupiranim podacima Zadržavamo H_0 .

#Jesu li neki žanrovi u prosjeku dulji od ostalih?

Kako bismo utvrdili razlikuju li se žanrovi po trajanju, uspoređujemo srednje vrijednosti duljine filma između više grupa (žanrova). Prvo koristimo Boxplot za vizualizaciju distribucije i Shapiro-Wilk test za provjeru normalnosti podataka. Budući da su varijance među žanrovima često nejednake (heteroskedastičnost), standardna ANOVA možda nije prikladna. Zato koristimo Leveneov test za provjeru jednakosti varijanci. Ako varijance nisu jednake, koristimo Welchov ANOVA ** koji je robusniji.

Hipoteze:

H_0 - Duljina filma ne ovisi o žanru filma H_1 - Duljina filma ovisi o žanru filma

```
## [1] "Tablica duljine trajanja filmova u žanrovima:"
```

```
## # A tibble: 14 x 3
##   'Main Genre' Avg_Duration Movie_Count
##   <chr>         <dbl>         <int>
## 1 Adventure     140.           45
## 2 Action        132.           39
## 3 Biography     132.          101
## 4 Western       130.            2
## 5 Drama         126.          215
## 6 Crime         122.           48
## 7 Horror        118.            3
## 8 Family        115            1
```

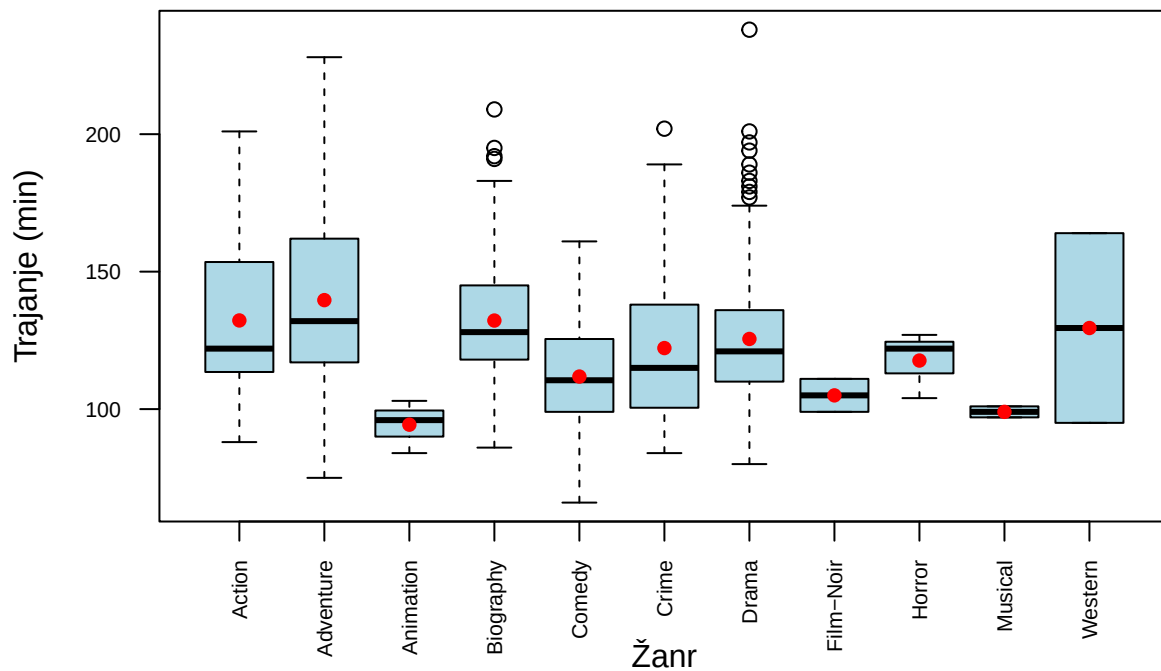


```
## 9 Comedy          112.          108
## 10 Film-Noir       105           2
## 11 Musical         99           2
## 12 Animation       94.3         3
## 13 Music           84           1
## 14 Romance         84           1
```

```
## [1] "Tablica duljine trajanja filmova u žanrovima nakon filtriranja:"
```

```
## # A tibble: 11 x 3
##   main_genre Avg_Duration Movie_Count
##   <fct>      <dbl>      <int>
## 1 Adventure  140.         45
## 2 Action    132.         39
## 3 Biography 132.        101
## 4 Western   130.          2
## 5 Drama     126.        215
## 6 Crime     122.         48
## 7 Horror    118.          3
## 8 Comedy    112.        108
## 9 Film-Noir 105           2
## 10 Musical   99           2
## 11 Animation 94.3          3
```

Trajanje filmova po žanrovima



Već u ovom prikazu možemo vidjeti razliku u trajanju filmova po žanrovima.

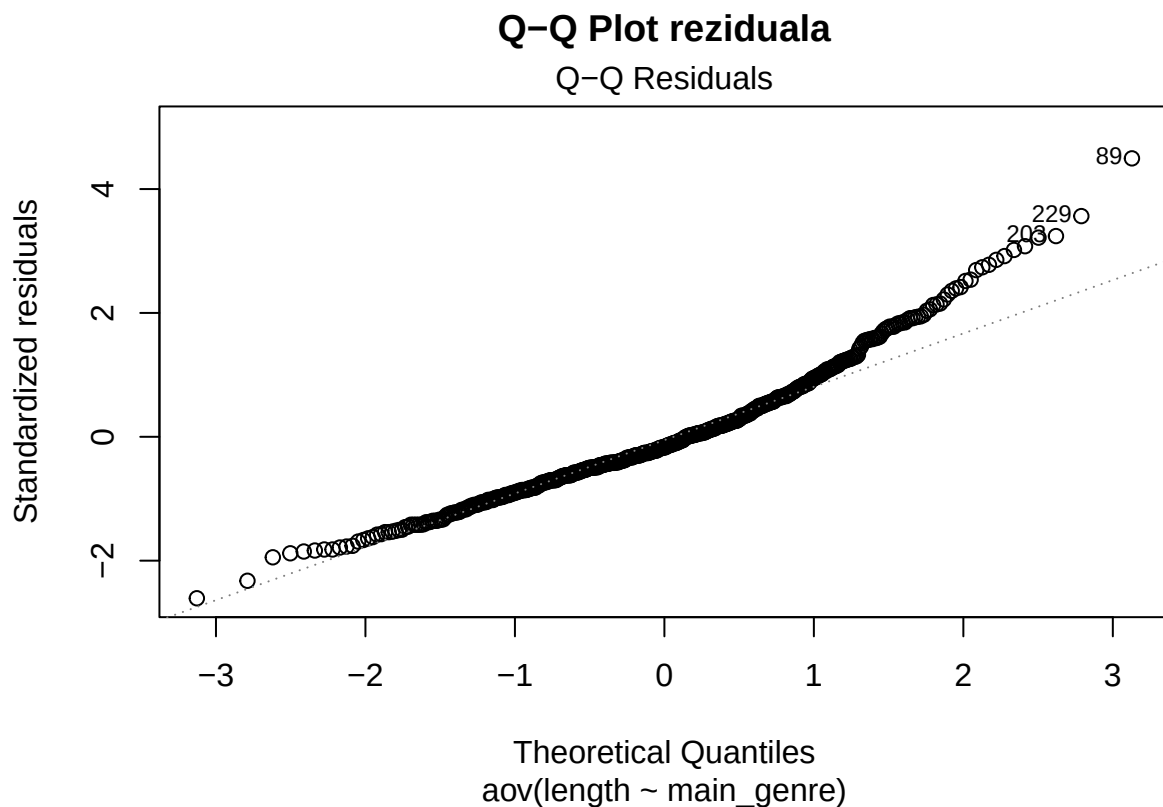
```
#Normalnost
print("Shapiro-Wilk test normalnosti:")
```

```
## [1] "Shapiro-Wilk test normalnosti:"
```

```
model <- aov(length ~ main_genre, data = data_clean)
shapiro_res <- shapiro.test(residuals(model))
print(shapiro_res)
```

```
##
## Shapiro-Wilk normality test
##
## data: residuals(model)
## W = 0.96046, p-value = 3.195e-11
```

```
# Q-Q Plot za normalnost
plot(model, which = 2, main = "Q-Q Plot reziduala")
```



Vizualni pregled Q-Q plota potvrđuje rezultate Shapiro-Wilk testa. Iako se većina podataka grupira oko teoretskog pravca normalnosti, postoje značajna odstupanja na krajevima distribucije. Ova odstupanja uzrokovana su ekstremnim vrijednostima (vrlo dugi filmovi), što je karakteristično za produkcije nominirane za nagradu Oscar. Zbog velikog uzorka, ova odstupanja ne ugrožavaju valjanost zaključka o razlikama među žanrovima.

```
#Homogenost
print("Test homogenosti varijanca (Leveneov test):")
```

```
## [1] "Test homogenosti varijanca (Leveneov test):"
```

```
levene_res <- leveneTest(length ~ main_genre, data = data_clean)
print(levene_res)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value    Pr(>F)
## group 10  2.8825 0.001613 **
##      557
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#ANOVA
print("ANOVA test:")
```

```
## [1] "ANOVA test:"
```

```
model <- aov(length ~ main_genre, data = data_clean)
levene_res <- leveneTest(length ~ main_genre, data = data_clean)

if(levene_res$`Pr(>F)`[1] > 0.05) {
  print("Standardni ANOVA test:") # Varijance su homogene -> Standardnu ANOVU
  print(summary(model))
} else {
  print("Welchov ANOVA test:") # Varijance nisu homogene (p < 0.05) -> Welchov ANOVA test
  welch_test <- oneway.test(length ~ main_genre, data = data_clean, var.equal = FALSE)
  print(welch_test)
}
```

```
## [1] "Welchov ANOVA test:"
##
## One-way analysis of means (not assuming equal variances)
##
## data: length and main_genre
## F = 13.214, num df = 10.000, denom df = 10.929, p-value = 9.722e-05
```

```
if(summary(model)[[1]][["Pr(>F)"]][1] < 0.05) {
  print("Post-hoc test (Tukey HSD):") # Usporedba parova žanrova
  tukey <- TukeyHSD(model)
  print(head(tukey$main_genre, 30)) # Prikaz prvih 30
}
```

```
## [1] "Post-hoc test (Tukey HSD):"
##
##      diff      lwr      upr      p adj
## Adventure-Action    7.38803419 -10.343156 25.1192245 9.600342e-01
## Animation-Action   -37.92307692 -86.481942 10.6357877 2.926458e-01
## Biography-Action   -0.03858847 -15.318043 15.2408656 1.000000e+00
```

```
## Comedy-Action      -20.41381766 -35.554704 -5.2729315 7.938218e-04
## Crime-Action       -10.04807692 -27.520113  7.4239590 7.438428e-01
## Drama-Action       -6.73082886 -20.836772  7.3751144 9.048552e-01
## Film-Noir-Action   -27.25641026 -86.016363 31.5035423 9.198480e-01
## Horror-Action      -14.58974359 -63.148608 33.9691210 9.966446e-01
## Musical-Action     -33.25641026 -92.016363 25.5035423 7.622404e-01
## Western-Action     -2.75641026 -61.516363 56.0035423 1.000000e+00
## Animation-Adventure -45.31111111 -93.638190  3.0159676 8.910902e-02
## Biography-Adventure -7.42662266 -21.952631  7.0993858 8.586382e-01
## Comedy-Adventure   -27.80185185 -42.182034 -13.4216696 4.550920e-08
## Crime-Adventure     -17.43611111 -34.253223 -0.6189994 3.466023e-02
## Drama-Adventure    -14.11886305 -27.404978 -0.8327484 2.649903e-02
## Film-Noir-Adventure -34.64444444 -93.212996 23.9241071 7.093750e-01
## Horror-Adventure    -21.97777778 -70.304857 26.3493009 9.291301e-01
## Musical-Adventure   -40.64444444 -99.212996 17.9241071 4.751518e-01
## Western-Adventure   -10.14444444 -68.712996 48.4241071 9.999751e-01
## Biography-Animation 37.88448845  -9.597857 85.3668340 2.623794e-01
## Comedy-Animation   17.50925926 -29.928678 64.9471962 9.831372e-01
## Crime-Animation     27.87500000 -20.357598 76.1075975 7.380145e-01
## Drama-Animation     31.19224806 -15.925573 78.3100691 5.491535e-01
## Film-Noir-Animation 10.66666667 -63.318761 84.6520939 9.999956e-01
## Horror-Animation    23.33333333 -42.841245 89.5079112 9.880440e-01
## Musical-Animation    4.66666667 -69.318761 78.6520939 1.000000e+00
## Western-Animation   35.16666667 -38.818761 109.1520939 9.070343e-01
## Comedy-Biography    -20.37522919 -31.593796 -9.1566621 4.089077e-07
## Crime-Biography     -10.00948845 -24.218002  4.1990248 4.512904e-01
## Drama-Biography     -6.69224039 -16.469124  3.0846430 4.969012e-01
```

P vrijednost u Leveneovom testu je $0.001613 < 0.05$ što znači da varijance nisu homogene što znači da trebamo koristiti Welchov ANOVA test.

U Welchovom ANOVA testu p-value = $9.722e-05$ što je znatno manje od 0.05 što znači da odbacujemo H_0 i zadržavamo H_1

Promatrajući p adj stupac u Post-hoc Tuckey HSD testu vidimo na primjer da je komedija znatno kraća od ostalih žanrova (p adj < 0.05)

Analiza sugerira da su drame i biografski filmovi u prosjeku dulji, dok su animirani filmovi i komedije značajno kraći, što ima smisla s obzirom na konvencije filmske industrije.

Kako su filmovi ocijenjeni s obzirom na njihovu starost?

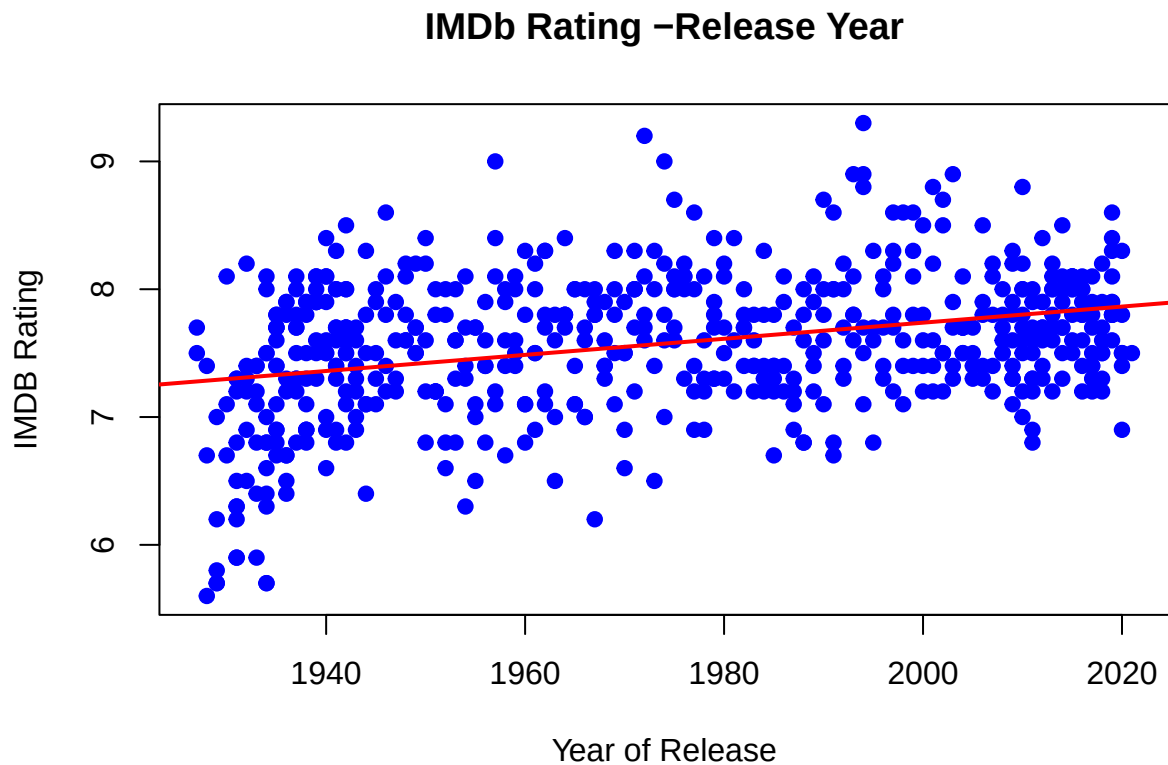
Ovdje ispitujemo povezanost između godine izlaska filma i njegove ocjene (IMDb, Tomato, Audience) koristeći linearnu regresiju. Cilj je vidjeti postoji li vremenski trend – jesu li stariji filmovi ili noviji filmovi bolje ocijenjeni. Prikazujemo Scatter plot s regresijskim pravcem te analiziramo Pearsonov koeficijent korelacije i R^2 vrijednost modela kako bismo utvrdili snagu te veze.

```
mean.imdb.rating <- mean(oscars_dataset2$imdb_rating)
mean.tomato.rating <- oscars_dataset2$tomato_rating %>% na.omit() %>% mean()
mean.audience.rating <- oscars_dataset2$audience_rating %>% na.omit() %>% mean()
mean.year <- mean(oscars_dataset2$y_release)
imdb_ocjena_starost <- select(oscars_dataset2, y_release, imdb_rating)
```

```

fit.imdb = lm(imdb_rating~y_release,data=imdb_ocjena_starost)
plot(imdb_ocjena_starost$y_release,imdb_ocjena_starost$imdb_rating,main = "IMDb Rating -Release Year",
     xlab = "Year of Release",
     ylab = "IMDb Rating",
     pch = 19,
     col = "blue")
#lines(imdb_ocjena_starost$y_release,fit.imdb$fitted.values,col='red')
abline(fit.imdb, col = "red", lwd = 2)

```

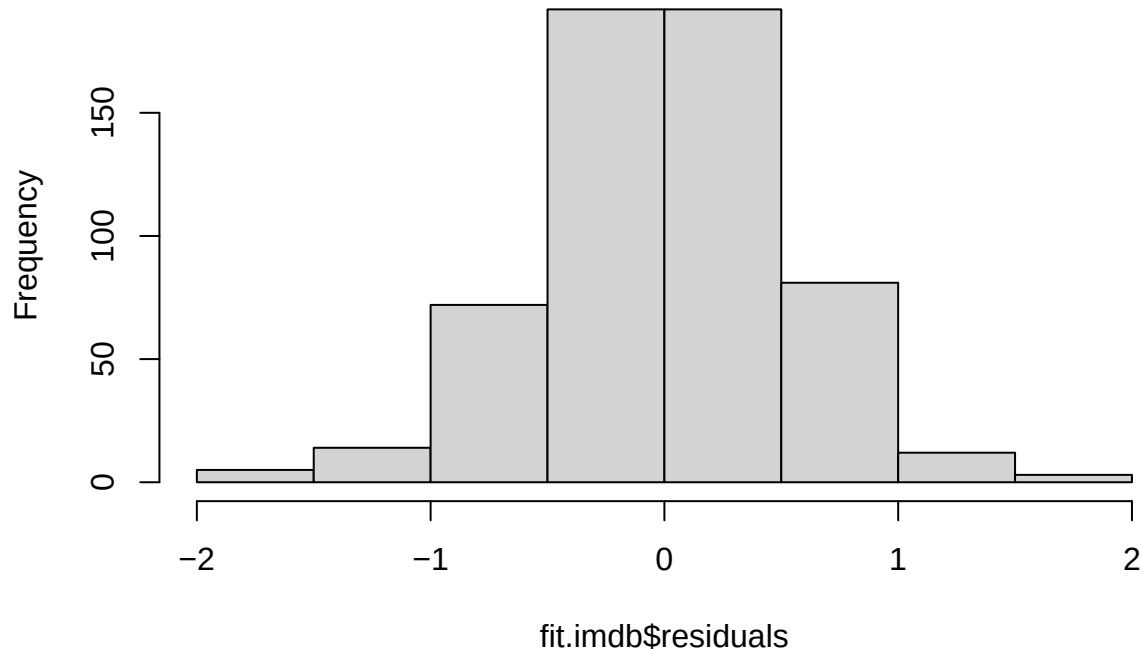


```

#raspodjela reziduala
hist(fit.imdb$residuals)

```

Histogram of fit.imdb\$residuals



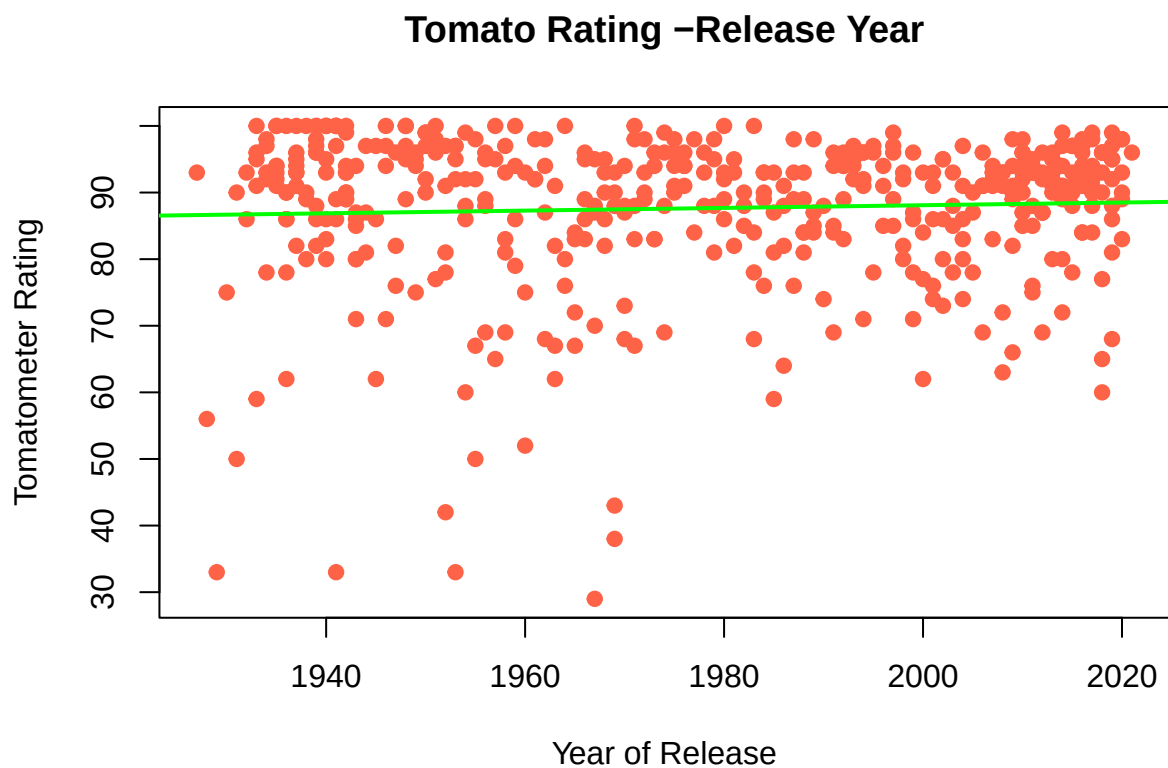
```
#studentova standardizacija reziduala  
#hist(rstandard(fit.imdb))  
#iscitavanje podataka uključujući prilagođeni R^2  
summary(fit.imdb)
```

```
##  
## Call:  
## lm(formula = imdb_rating ~ y_release, data = imdb_ocjena_starost)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -1.68474 -0.35157  0.00511  0.36096  1.63815   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept) -4.8580725   1.4910503   -3.258  0.00119 **   
## y_release    0.0062981   0.0007555    8.336 5.78e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 0.5288 on 569 degrees of freedom  
## Multiple R-squared:  0.1088, Adjusted R-squared:  0.1073   
## F-statistic: 69.49 on 1 and 569 DF, p-value: 5.783e-16
```

```
#korelacija  
cor(oscars_dataset2$y_release,oscars_dataset2$imdb_rating)
```

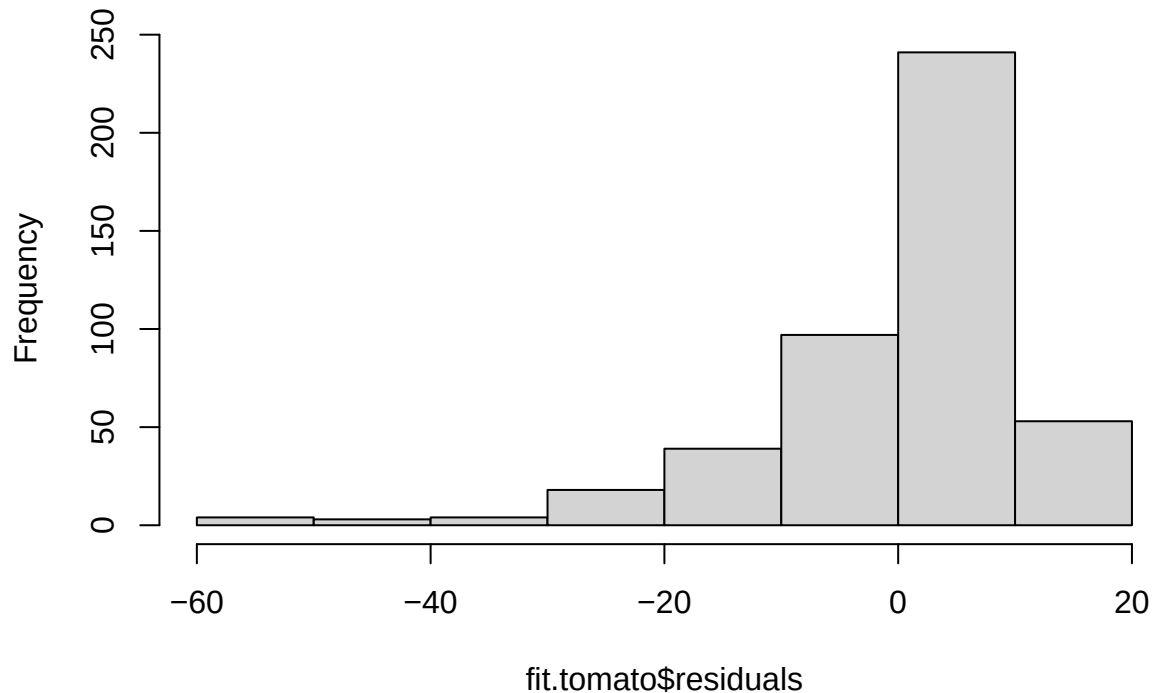
```
## [1] 0.3299097
```

```
fit.tomato = lm(tomato_rating~y_release,data=oscars_dataset2)  
plot(oscars_dataset2$y_release,oscars_dataset2$tomato_rating,main = "Tomato Rating -Release Year",  
      xlab = "Year of Release",  
      ylab = "Tomatometer Rating",  
      pch = 19,  
      col = "tomato")  
abline(fit.tomato, col = "green", lwd = 2)
```



```
hist(fit.tomato$residuals)
```

Histogram of fit.tomato\$residuals



```
#hist(rstandard(fit.tomato))
summary(fit.tomato)
```

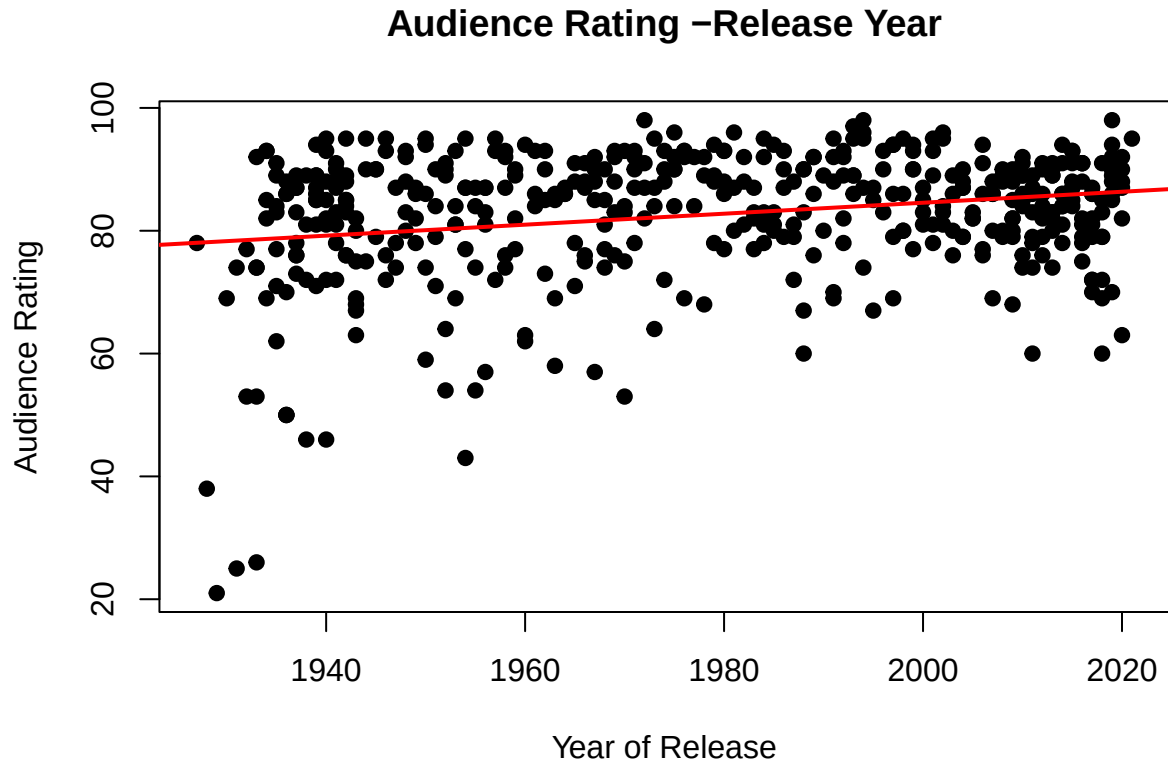
```
##
## Call:
## lm(formula = tomato_rating ~ y_release, data = oscars_dataset2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -58.427  -3.897   3.079   7.546  13.262
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  47.55938   37.29628   1.275   0.203
## y_release     0.02027    0.01886   1.075   0.283
##
## Residual standard error: 11.53 on 457 degrees of freedom
## (112 observations deleted due to missingness)
## Multiple R-squared:  0.002522,    Adjusted R-squared:  0.0003392
## F-statistic: 1.155 on 1 and 457 DF,  p-value: 0.283
```

```
cor(oscars_dataset2$y_release,oscars_dataset2$imdb_rating)
```

```
## [1] 0.3299097
```

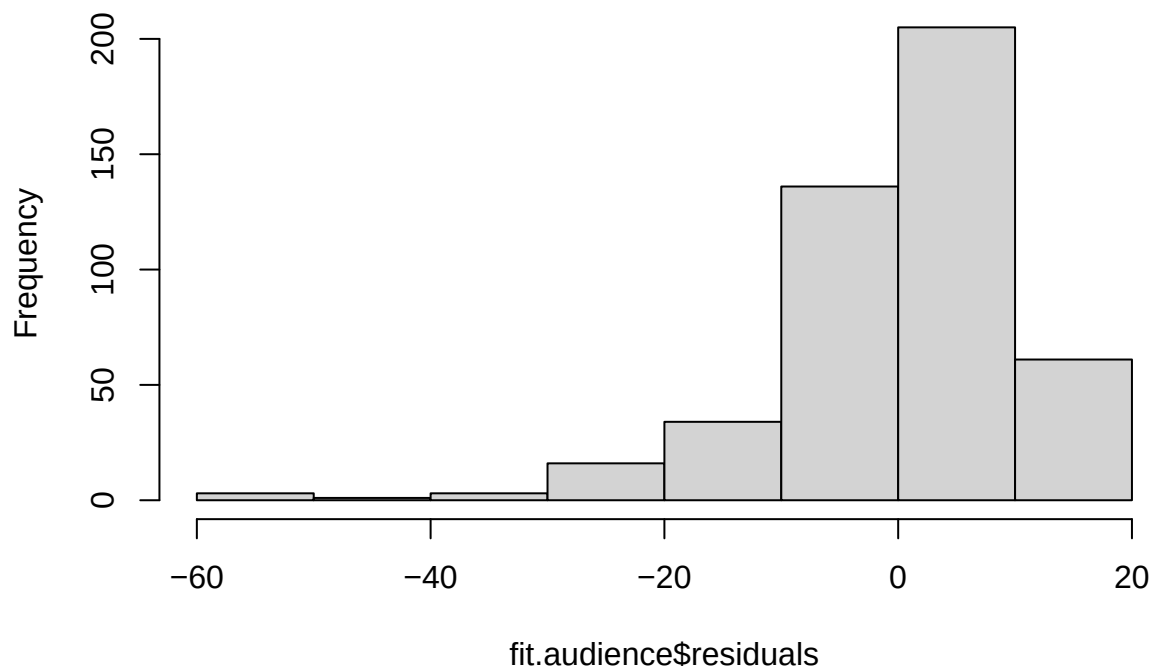


```
fit.audience = lm(audience_rating~y_release,data=oscars_dataset2)
plot(oscars_dataset2$y_release,oscars_dataset2$audience_rating,main = "Audience Rating -Release Year",
     xlab = "Year of Release",
     ylab = "Audience Rating",
     pch = 19,
     col = "black")
abline(fit.audience, col = "red", lwd = 2)
```



```
hist(fit.audience$residuals)
```

Histogram of fit.audience\$residuals



```
#hist(rstandard(fit.audience))
summary(fit.audience)
```

```
##
## Call:
## lm(formula = audience_rating ~ y_release, data = oscars_dataset2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -57.218  -4.694   2.008   7.097  15.940
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -94.14921   34.26520  -2.748  0.00624 **
## y_release      0.08936    0.01732   5.158 3.73e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.59 on 457 degrees of freedom
## (112 observations deleted due to missingness)
## Multiple R-squared:  0.05501,    Adjusted R-squared:  0.05295
## F-statistic: 26.61 on 1 and 457 DF,  p-value: 3.726e-07
```

Ovakvo oblikova rezidualni govore u korist uvjerljivosti našeg linearnog modela. Očitujemo malu vrijednost R^2 čime dalje potvrđujemo tvrdnju da starost filma nema utjecaj na ocjenu.

Moguće je na prvi pogled zaključiti kako godina izdanja ne utječe mnogo na ocjenu filma. Mali pozitivan nagib je upečaćen no ne smatramo ga toliko signifikantnim. Najveći nagib je u IMDB grafu.

```
# test normalnosti reziduala p < 0.05  
shapiro.test(fit.imdb$residuals)
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: fit.imdb$residuals  
## W = 0.99448, p-value = 0.03708
```

```
shapiro.test(fit.tomato$residuals)
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: fit.tomato$residuals  
## W = 0.82161, p-value < 2.2e-16
```

```
shapiro.test(fit.audience$residuals)
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: fit.audience$residuals  
## W = 0.88293, p-value < 2.2e-16
```

```
# p vrijednost je manje od 0.05 za sve a možemo zaključiti normalnost
```

```
cor.test(oscars_dataset2$y_release, oscars_dataset2$imdb_rating,method = "pearson")
```

```
##  
## Pearson's product-moment correlation  
##  
## data: oscars_dataset2$y_release and oscars_dataset2$imdb_rating  
## t = 8.3363, df = 569, p-value = 5.783e-16  
## alternative hypothesis: true correlation is not equal to 0  
## 95 percent confidence interval:  
## 0.2547525 0.4011050  
## sample estimates:  
## cor  
## 0.3299097
```

```
cor.test(oscars_dataset2$y_release, oscars_dataset2$tomato_rating,method = "pearson")
```

```
##  
## Pearson's product-moment correlation  
##  
## data: oscars_dataset2$y_release and oscars_dataset2$tomato_rating
```

```
## t = 1.0749, df = 457, p-value = 0.283
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.04149945 0.14109647
## sample estimates:
##      cor
## 0.05021814
```

```
cor.test(oscars_dataset2$y_release, oscars_dataset2$audience_rating, method = "pearson")
```

```
##
## Pearson's product-moment correlation
##
## data:  oscars_dataset2$y_release and oscars_dataset2$audience_rating
## t = 5.158, df = 457, p-value = 3.726e-07
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.1461621 0.3192250
## sample estimates:
##      cor
## 0.2345512
```

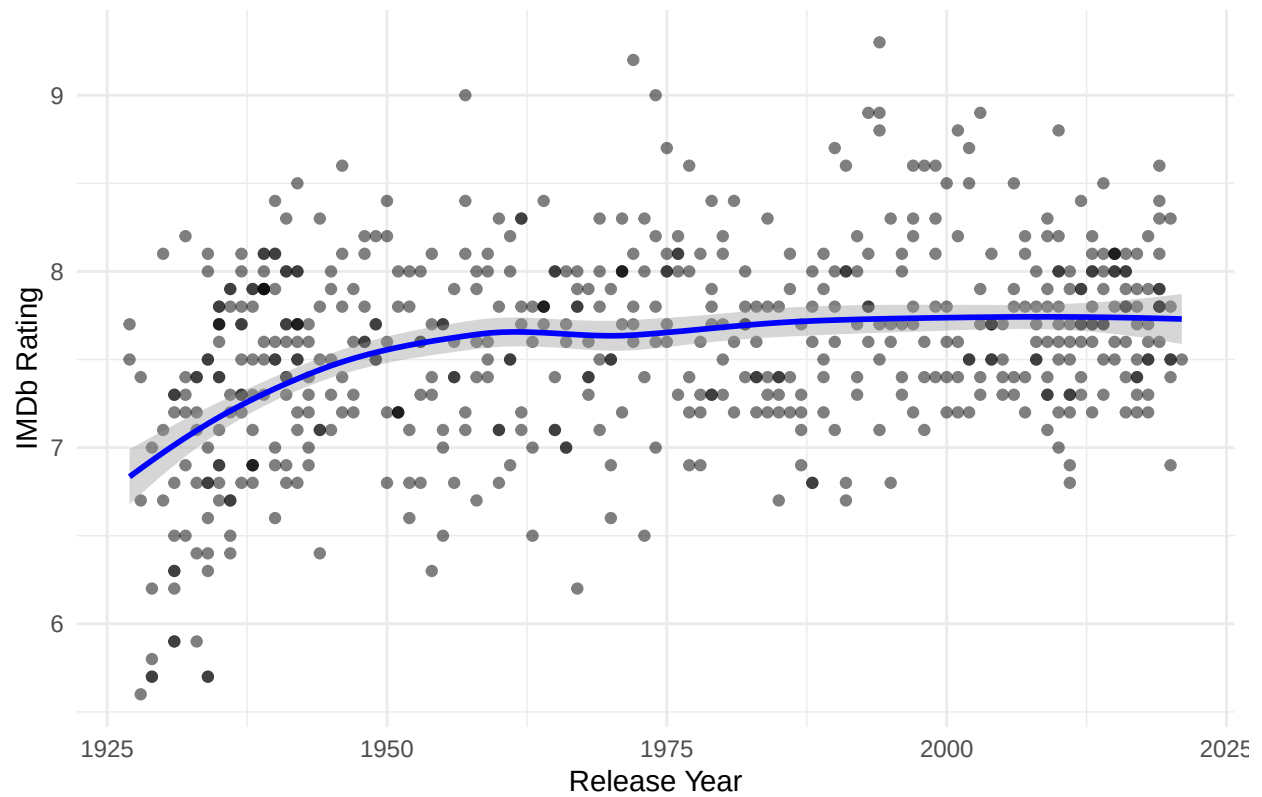
Nemožemo odbaciti hipotezu da su reziduali normalno distribuirani.

Pearson korelacijskim testom možemo vidjeti da su generalno slabo korelirani stime da imdb ima najveću korelaciju sa 0.33 što se može smatrati umjerenom korelacijom. Kako su sve tri umjerene no pozitivne postoji mogućnost da noviji filmovi imaju bolje ocijene.

Kaka linearna regresija ne daje čvrst zaključak pokušamo LOESS metodom

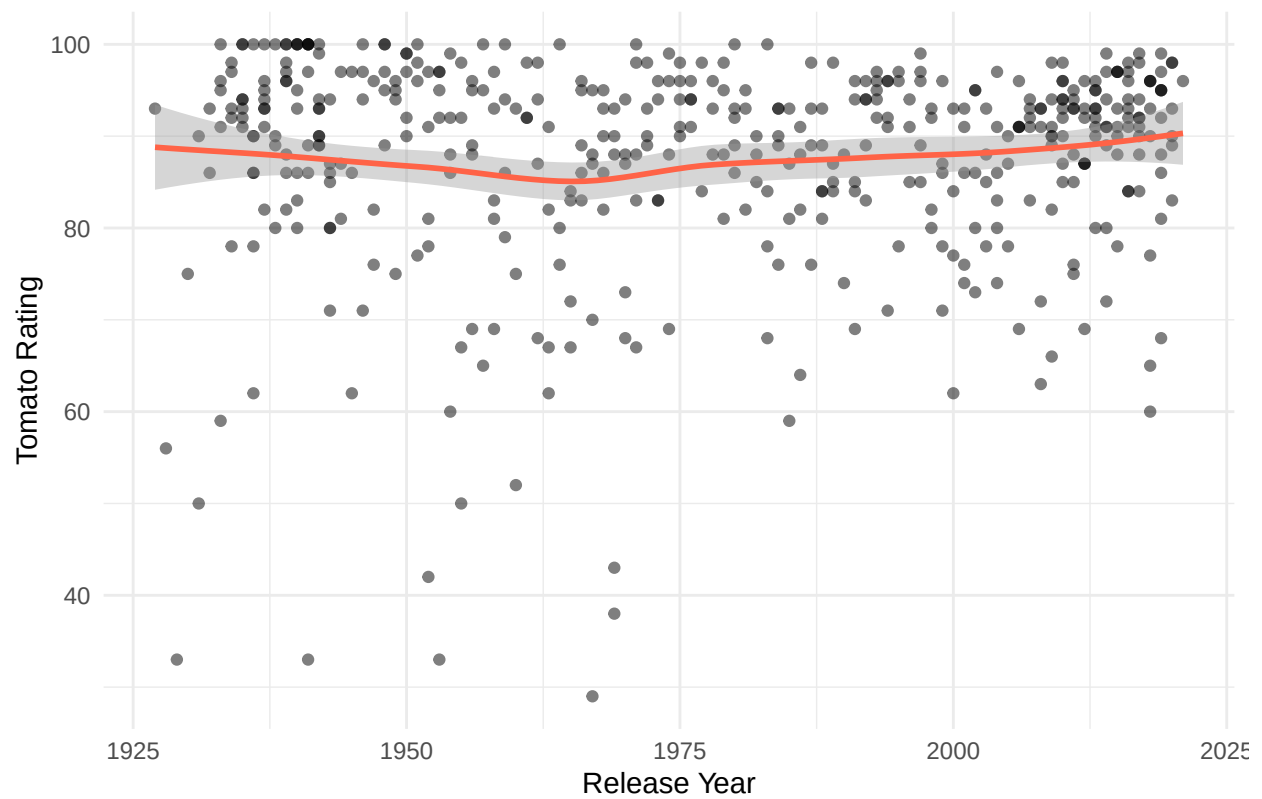
```
ggplot(imdb_ocjena_starost, aes(x = y_release, y = imdb_rating)) +
  geom_point(alpha = 0.5) + # alpha makes points semi-transparent to see density
  geom_smooth(method = "loess", color = "blue", se = TRUE) + # "lm" = linear model
  labs(title = "Relationship Between Movie Age and IMDb Rating",
        x = "Release Year",
        y = "IMDb Rating") +
  theme_minimal()
```

Relationship Between Movie Age and IMDb Rating

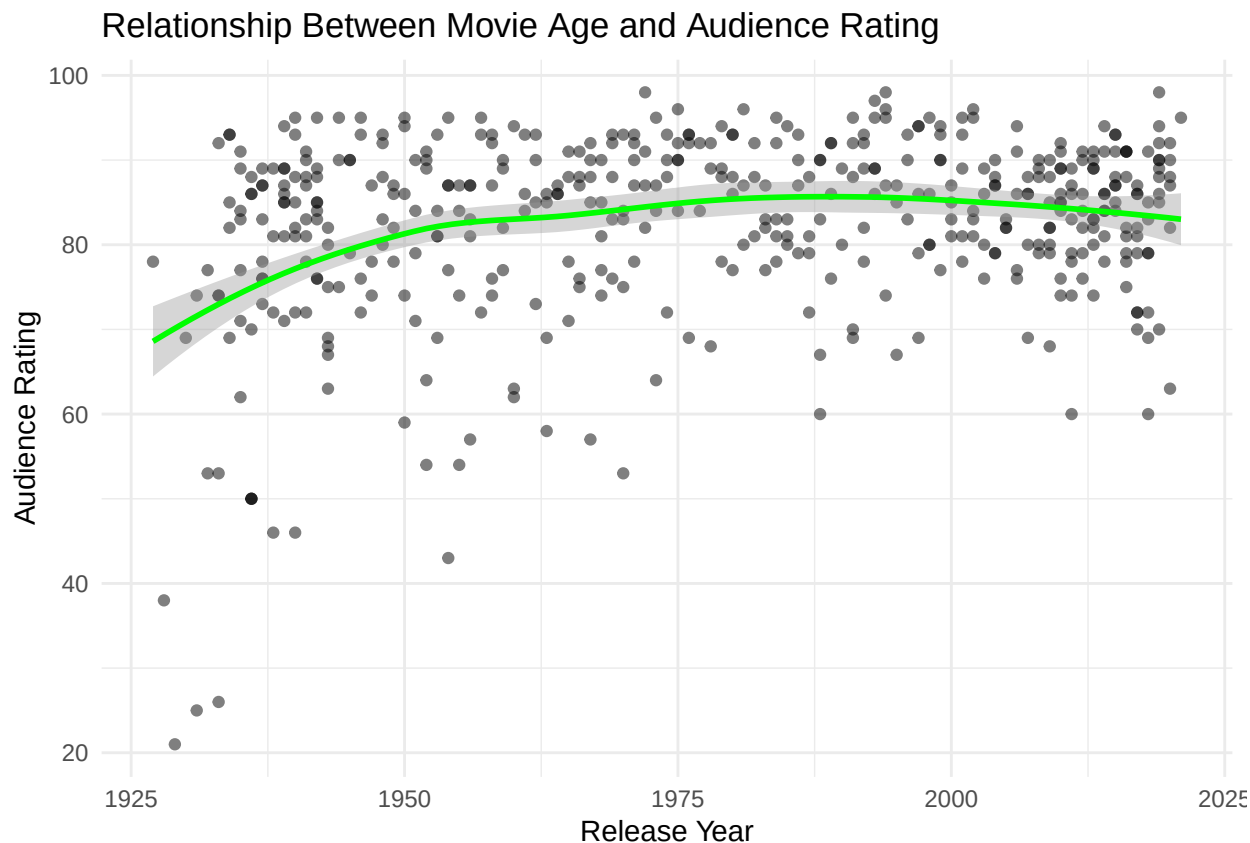


```
ggplot(oscars_dataset2, aes(x = y_release, y = tomato_rating)) +  
  geom_point(alpha = 0.5) +  
  geom_smooth(method = "loess", color = "tomato", se = TRUE) +  
  labs(title = "Relationship Between Movie Age and Tomatometer Rating",  
        x = "Release Year",  
        y = "Tomato Rating") +  
  theme_minimal()
```

Relationship Between Movie Age and Tomatometer Rating



```
ggplot(oscars_dataset2, aes(x = y_release, y = audience_rating)) +  
  geom_point(alpha = 0.5) +  
  geom_smooth(method = "loess", color = "green", se = TRUE) +  
  labs(title = "Relationship Between Movie Age and Audience Rating",  
        x = "Release Year",  
        y = "Audience Rating") +  
  theme_minimal()
```



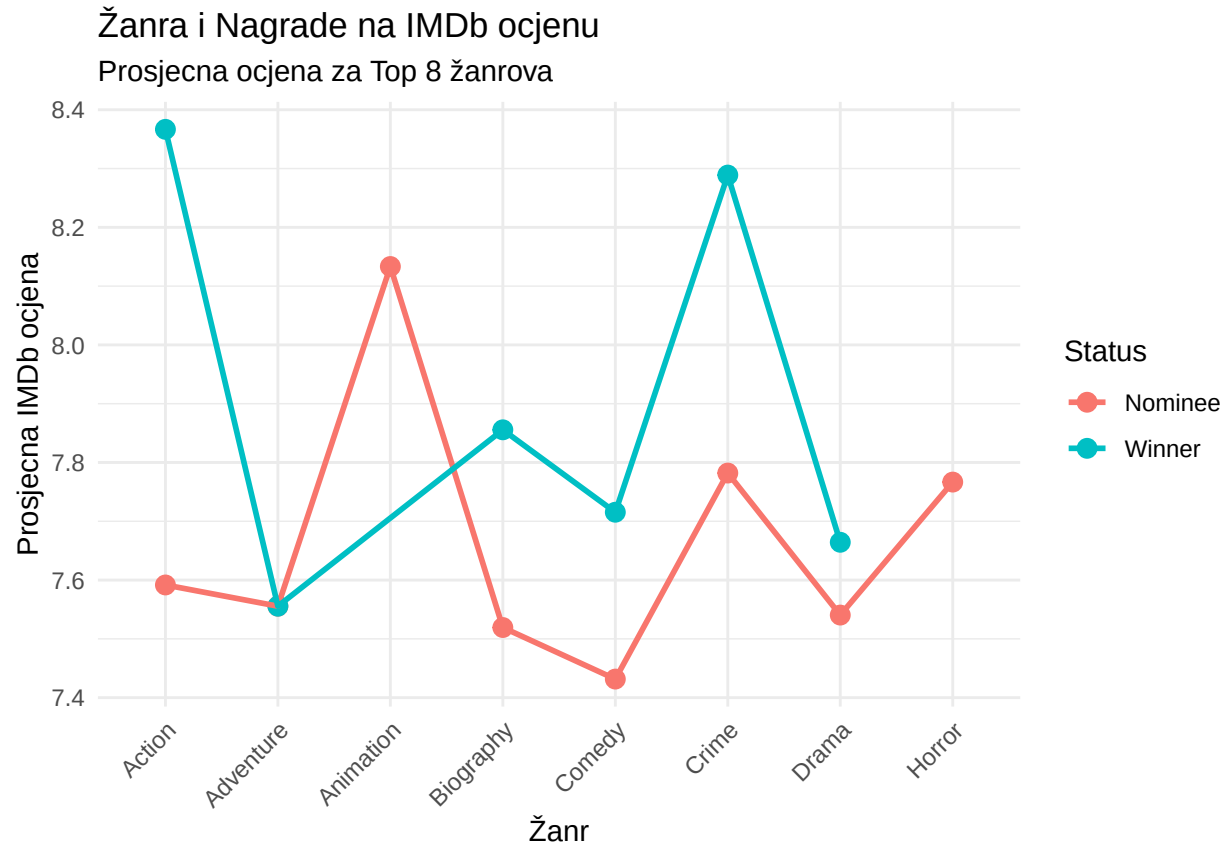
Ovaj model nam govori više u korist pretpostavke da su stariji filmovi lošije ocijenjeni (pogotvo IMDb i Audience ratings).

Vizualni i statistički rezultati (vrlo nizak R^2) pokazuju da godina izlaska filma nije dobar prediktor njegove kvalitete. Iako postoji blaga korelacija kod IMDb ocjena, ona nije dovoljno snažna da bismo zaključili kako starost filma presudno utječe na ocjenu. Reziduali su uglavnom normalno distribuirani, što validira model, ali sam nagib pravca je zanemariv.

Možemo li naslutiti kako je film ocijenjen pomoću drugih značajki?

Kako bismo pronašli faktore koji utječu na ocjenu, prvo koristimo korelacijsku matricu za numeričke varijable (npr. trajanje, broj glasova). Zatim analiziramo kvalitativne značajke (poput statusa pobjednika Oscara) koristeći Wilcoxon test (jer podaci nisu normalno distribuirani) kako bismo vidjeli imaju li pobjednici značajno više ocjene. Na kraju, koristimo Bootstrap metodu za procjenu intervala pouzdanosti prosječne ocjene, što nam omogućuje da identificiramo “vrhunske” producente čiji filmovi statistički odskakuju od prosjeka.

Možemo iskoristiti nekoliko metoda kako bi vidjeli možemo li koristiti druge merike da bi mogli naslutiti ocjenu filma.



Vizualno odmah možemo naslutiti da pobjednici dobivaju bolje ocjene

Numerička analiza

```
# Biramo samo numeričke stupce za nalizu
numerical_data <- oscar_dataset2[, c("length", "imdb_rating", "imdb_votes",
                                     "tomato_rating", "tomato_count", "audience_rating", "audience_count")]

# Kalkuliramo matrix korelacije koji nam je pogod
cor_matrix <- cor(numerical_data, use = "complete.obs")
print(cor_matrix)
```

```
##           length imdb_rating imdb_votes tomato_rating tomato_count
## length      1.00000000  0.2334096  0.26604443   -0.18968511  0.07762022
## imdb_rating  0.23340962  1.00000000  0.61672439    0.46632079  0.24633971
## imdb_votes   0.26604443  0.6167244  1.00000000    0.09744051  0.47035921
## tomato_rating -0.18968511  0.4663208  0.09744051    1.00000000  0.12644534
## tomato_count  0.07762022  0.2463397  0.47035921    0.12644534  1.00000000
## audience_rating 0.13647250  0.8485251  0.33591478    0.50462680  0.15622655
## audience_count 0.18911348  0.1679893  0.31230476   -0.00915894  0.04615373
##
## audience_rating audience_count
## length          0.13647250    0.18911348
## imdb_rating      0.84852507    0.16798930
## imdb_votes       0.33591478    0.31230476
## tomato_rating    0.50462680   -0.00915894
## tomato_count     0.15622655    0.04615373
## audience_rating  1.00000000   -0.01560293
```



```
## audience_count      -0.01560293      1.00000000
```

Ne primjećujemo znatne korelacije koje bi nam mogle korisiti u predikciji
Moramo analizirati kvalitativne stupce.

```
genre_analysis <- oscars_dataset2 %>%
  group_by(main_genre) %>%
  summarise(
    avg_rating = mean(imdb_rating, na.rm = TRUE),
    movie_count = n()
  ) %>%
  arrange(desc(avg_rating))

print(genre_analysis)
```

```
## # A tibble: 14 x 3
##   main_genre avg_rating movie_count
##   <chr>      <dbl>      <int>
## 1 Animation      8.13          3
## 2 Crime          7.88         48
## 3 Family         7.8           1
## 4 Horror         7.77          3
## 5 Action         7.65         39
## 6 Biography      7.58        101
## 7 Drama          7.56        215
## 8 Adventure      7.56          45
## 9 Comedy         7.47        108
## 10 Film-Noir     7.45           2
## 11 Romance       6.7            1
## 12 Western       6.35           2
## 13 Musical       6.25           2
## 14 Music         5.7            1
```

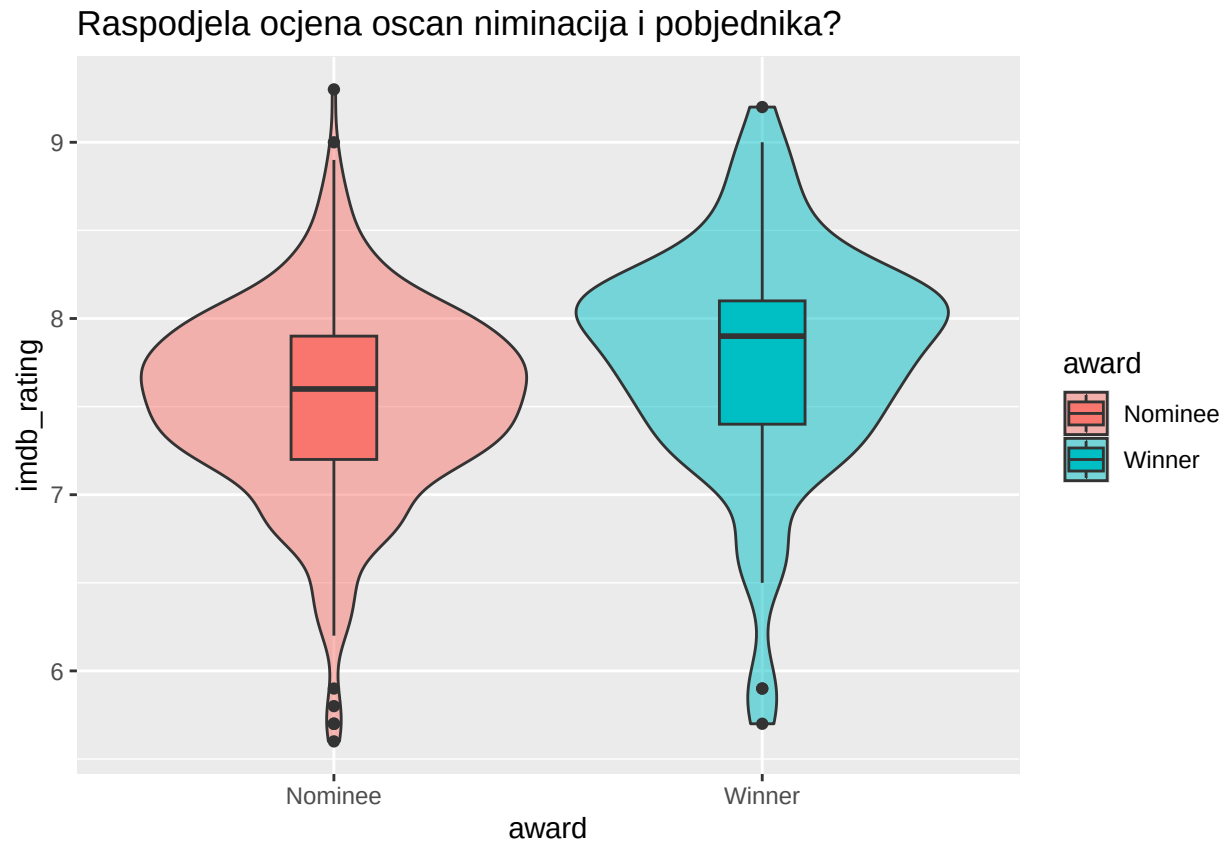
Možemo se i zapitati imaju li Oscar pobjednici bolje ocjene.

```
imdb_stats <- oscars_dataset %>%
  group_by(Award) %>%
  summarise(
    Prosjek_IMDb = mean(`IMDB Rating`, na.rm = TRUE),
    Medijan_IMDb = median(`IMDB Rating`, na.rm = TRUE),
    Broj_filmova = n()
  )

print(imdb_stats)
```

```
## # A tibble: 2 x 4
##   Award   Prosjek_IMDb Medijan_IMDb Broj_filmova
##   <chr>      <dbl>      <dbl>      <int>
## 1 Nominee      7.53        7.6        477
## 2 Winner       7.78        7.9         94
```

```
#Vidimo da ima vidljive razlike u prosjeku i u medijanu
ggplot(oscars_dataset2, aes(x = award, y = imdb_rating, fill = award)) +
  geom_violin(alpha = 0.5) +
  geom_boxplot(width = 0.2) +
  labs(title = "Raspodjela ocjena oscan niminacija i pobjednika?")
```



```
# Pomoću t testa možemo testirati jesu li prosjeci ova dva skupa stvarno statistički različita
# Znamo da su dva skupa nezavisna ner ne dijele istu značajku
#Koristi ćemo t test pa prvo se moramo uvjeriti normalnost podataka"
winners <- oscars_dataset2 %>% filter(award == "Winner") %>% select(imdb_rating) %>% pull(imdb_rating)
shapiro.test(winners)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  winners
## W = 0.95032, p-value = 0.001322
```

```
nominees <- oscars_dataset2 %>% filter(award == "Nominee") %>% select(imdb_rating) %>% pull(imdb_rating)
shapiro.test(nominees)
```

```
##
##  Shapiro-Wilk normality test
##
```

```
## data:  nominees
## W = 0.97874, p-value = 1.939e-06
```

```
# pošto je  $p < 0.01$  za oba testa možemo zaključiti da nisu normalno distribuirani
#Moramo koristiti ne parametarsku metodu
wilcox.test(winners, nominees, alternative = "greater")
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data:  winners and nominees
## W = 28913, p-value = 4.294e-06
## alternative hypothesis: true location shift is greater than 0
```

```
#  $p < 0.0001$  možemo gotovo sigurno odbaciti hipotezu u korist toga da oscar pobjednici imaju bolji rating
```

Biti pobjednik oscara jedan je od prediktora višeg ratinga

Možemo pretpostaviti da neki producenti proizvode bolje filmove

```
#Kako bi mogli vidjeti koji studiji proizvode visoko ocjenjene filmove moramo znati interval pouzdanosti
#Koristimo bootstrap metodu jer imdb rating nisu normalno distribuirani i robusno je
```

```
# Funkcija koja računa prosjek (potrebna za bootstrap)
boot_mean <- function(data, indices) {
  return(mean(data[indices], na.rm = TRUE))
}
```

```
# Pokretanje bootstrapa (1000 simulacija)
set.seed(553343) # Za ponovljivost
boot_results <- boot(data = oscars_dataset2$imdb_rating, statistic = boot_mean, R = 1000)
```

```
# Dobivanje 95% intervala pouzdanosti
ci <- boot.ci(boot_results, conf = 0.99, type = "perc")
#perc type Directly uses quantiles from the bootstrap distribution as interval endpoints (e.g., the 2.5
#types valjda treba ili perc ili basic
```

```
gornja_granica <- ci$percent[5]
```

```
#Sve prodksijske kompanije koje su iznad gornje granice možemo poprilično sigurno reći da proizvode vis
```

```
uspjesni_producenti <- oscars_dataset2 %>%
  group_by(producer) %>%
  summarise(
    prosjek_producenta = mean(imdb_rating, na.rm = TRUE),
    broj_filmova = n()
  ) %>%
  filter(broj_filmova >= 3) %>%
  filter(prosjek_producenta > gornja_granica) %>%
  arrange(desc(prosjek_producenta))

print(uspjesni_producenti)
```

```
## # A tibble: 7 x 3
```

##	producer	prosjek_producenta	broj_filmova
##	<chr>	<dbl>	<int>
## 1	Lawrence Bender	8.5	3
## 2	Stanley Kubrick	8.27	3
## 3	Irwin Winkler and Robert Chartoff	8.03	3
## 4	Sam Spiegel	7.92	4
## 5	Stanley Kramer	7.73	6
## 6	Robert Wise	7.7	3
## 7	Selznick International Pictures	7.7	5

Zaključak: Na temelju numeričke korelacijske analize i kvalitativnih usporedbi ne pronalazimo jednu dominantnu značajku koja sama pouzdano objašnjava ocjenu filma. Ipak, status pobjednika i pojedini producenti/žanrovi signaliziraju viši prosjek ocjena, što sugerira da kombinacija kategorijskih značajki daje smisleniji uvid od samih numeričkih metrike. Zaključujemo da su ovih 7 producenata proizvode statistički bolje filmove po tome što imaju više filmova (znači da imamo barem neku populaciju za proučiti) i prosječna ocjena njihovih filmova je veća od gornje granice intervala pouzdanosti za $\alpha = 0.01$ što znači da su određnu producenti znalaka kvalitete

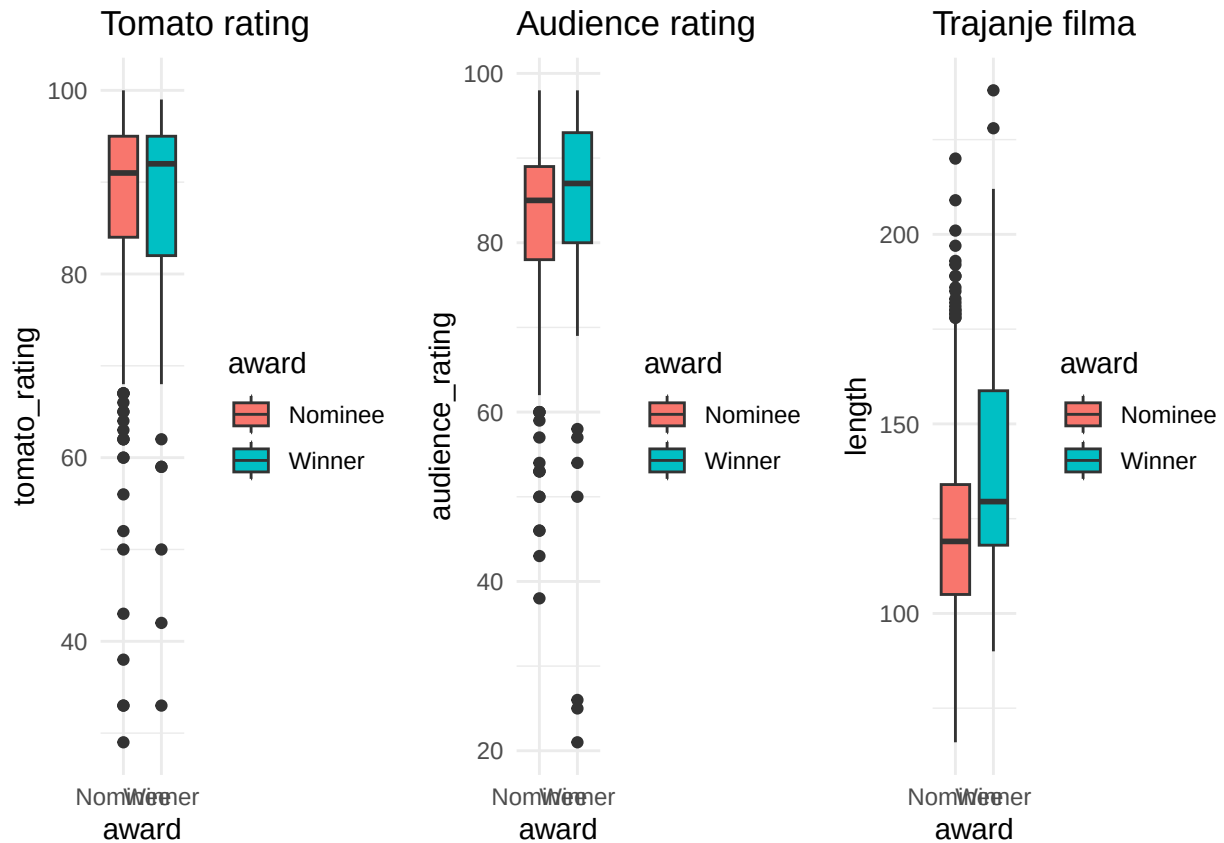
#Možemo li pomoću dostupnih značajki pouzdano razlikovati pobjednike od ostalih?

```
p1 <- ggplot(oscars_dataset2, aes(x = award, y = tomato_rating, fill = award)) +
  geom_boxplot() +
  labs(title = "Tomato rating ") + theme_minimal()

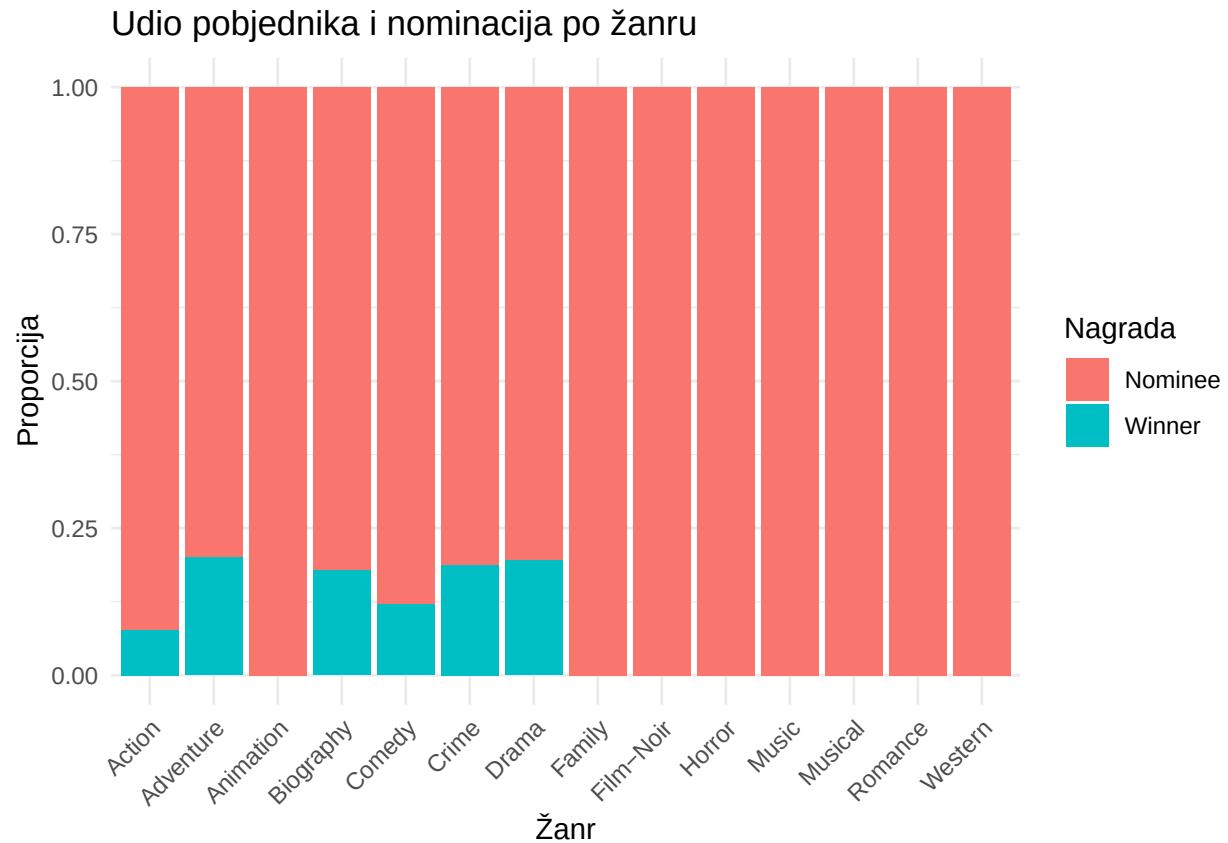
p2 <- ggplot(oscars_dataset2, aes(x = award, y = length, fill = award)) +
  geom_boxplot() +
  labs(title = "Trajanje filma") + theme_minimal()

p3 <- ggplot(oscars_dataset2, aes(x = award, y = audience_rating, fill = award)) +
  geom_boxplot() +
  labs(title = "Audience rating") + theme_minimal()

grid.arrange(p1, p3, p2, ncol = 3)
```



```
#Usporedba distribucija po žanru (stacked bar)
genre_award_counts <- oscars_dataset2 %>%
  count(main_genre, award) %>%
  group_by(main_genre) %>%
  mutate(prop = n / sum(n)) %>%
  ungroup()
ggplot(genre_award_counts, aes(x = main_genre, y = prop, fill = award)) +
  geom_col() +
  labs(
    title = "Udio pobjednika i nominacija po žanru",
    x = "Žanr",
    y = "Proporcija",
    fill = "Nagrada"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Stacked bar prikaz žanrova daje kontekst za moguće razlike u sastavu nominacija i pobjeda po žanru. To pomaže povezati numeričke rezultate s kvalitativnim karakteristikama.

```
# Logistička regresija
logit_data <- oscars_dataset2 %>%
  mutate(award_binary = ifelse(award == "Winner", 1, 0)) %>%
  select(award_binary, imdb_rating, tomato_rating, audience_rating, length, imdb_votes, main_genre) %>%
  drop_na()

logit_model <- glm(
  award_binary ~ imdb_rating + tomato_rating + audience_rating + length + imdb_votes + main_genre,
  data = logit_data,
  family = binomial
)

summary(logit_model)
```

```
##
## Call:
## glm(formula = award_binary ~ imdb_rating + tomato_rating + audience_rating +
##      length + imdb_votes + main_genre, family = binomial, data = logit_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -8.002e+00  3.320e+00  -2.411   0.0159 *
## imdb_rating    8.576e-01  6.690e-01   1.282   0.1998
```

```
## tomato_rating      -1.421e-03  1.343e-02 -0.106  0.9157
## audience_rating    -3.905e-02  2.442e-02 -1.599  0.1098
## length             1.256e-02  5.277e-03  2.380  0.0173 *
## imdb_votes         6.857e-07  5.002e-07  1.371  0.1704
## main_genreAdventure 1.524e+00  8.070e-01  1.888  0.0590 .
## main_genreAnimation -1.359e+01  1.365e+03 -0.010  0.9921
## main_genreBiography 1.609e+00  7.439e-01  2.163  0.0305 *
## main_genreComedy    1.499e+00  7.773e-01  1.929  0.0537 .
## main_genreCrime     1.429e+00  7.930e-01  1.802  0.0716 .
## main_genreDrama     1.640e+00  7.205e-01  2.276  0.0228 *
## main_genreFilm-Noir -1.299e+01  2.400e+03 -0.005  0.9957
## main_genreHorror    -1.353e+01  1.691e+03 -0.008  0.9936
## main_genreWestern   -1.300e+01  2.400e+03 -0.005  0.9957
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 439.87  on 458  degrees of freedom
## Residual deviance: 408.07  on 444  degrees of freedom
## AIC: 438.07
##
## Number of Fisher Scoring iterations: 15
```

```
logit_null <- glm(award_binary ~ 1, data = logit_data, family = binomial)
mcfadden_r2 <- 1 - (logLik(logit_model) / logLik(logit_null))
print(paste("McFadden pseudo-R^2:", round(as.numeric(mcfadden_r2), 3)))
```

```
## [1] "McFadden pseudo-R^2: 0.072"
```

Logistički model pokazuje da su duljina filma** i određeni žanrovi (Drama,Biografija,Avantura itd.) statistički značajni prediktori pobjede. Međutim, nizak McFadden pseudo-R² (0.072) sugerira da samo ove varijable nisu dovoljne za pouzdano predviđanje pobjednika. Iako pobjednici imaju statistički različite distribucije duljine i IMDb ocjena u odnosu na nominirane, preklapanje je preveliko da bi se pobjednik mogao deterministički odvojiti samo na temelju ovih podataka.

```
# Statistički testovi - Wilcoxon rank-sum
# H0: Nema razlike u distribuciji između pobjednika i nominiranih
test_imdb <- wilcox.test(imdb_rating ~ award, data = oscars_dataset2)
test_len <- wilcox.test(length ~ award, data = oscars_dataset2)
test_tomato <- wilcox.test(tomato_rating ~ award, data = oscars_dataset2)

print(paste("P-vrijednost za IMDb:", test_imdb$p.value))
```

```
## [1] "P-vrijednost za IMDb: 8.58788614504331e-06"
```

```
print(paste("P-vrijednost za Duljinu:", test_len$p.value))
```

```
## [1] "P-vrijednost za Duljinu: 6.71961208650246e-06"
```

```
print(paste("P-vrijednost za Tomato:", test_tomato$p.value))
```

```
## [1] "P-vrijednost za Tomato: 0.91513884267567"
```

Wilcoxon testovi pokazuju da postoje razlike u raspodjelama IMDb ocjena i duljine između pobjednika i nominiranih (p-vrijednosti su vrlo male), dok Tomato ocjene ne pokazuju statistički značajnu razliku ($p = 0.915$).

Možemo detektirati obrasce i statističke razlike, ali model nema dovoljno snage da se pobjednici pouzdano razlikuju od nominiranih samo na temelju dostupnih značajki.